CE3420

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Assignment
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Variables

In [1]:
```
""" Question
We have a variable named distance_km below with the value 4135 - indicati
Create another variable called distance_mi and store the distance value i

Hint1: 1 mile = 1.60934 kilometers
Add the code in the cell below and run it. The output should be 2569.37
"""

distance_km = 4135
# Remove this line and add code here
distance_mi = distance_km / 1.60934

print(distance_mi)
```

```
2569.376266046951
```

Data Structures

In [2]:
```
""" Question
From the dictionary below, how do you access the latitude and longitude v
print the latitude and longitude of new york city by extracting it from t
The expected output should look like below.
40.661
-73.944
"""

nyc_data = {'city': 'New York', 'population': 8175133, 'coordinates': (40

print(nyc_data['coordinates'][0])
print(nyc_data['coordinates'][1])
```

```
40.661
-73.944
```

String Operations

In [3]:
```
""" Question
Use the string slicing to extract and print the degrees, minutes and seco
The output should be as follows
37
46
26.2992
"""
```

```python
latitude = '''37° 46' 26.2992"'''

deg = latitude[:latitude.index('°')]
minu = latitude[latitude.index('°') + 2 : latitude.index("'")]
sec = latitude[latitude.index("'") + 2 : latitude.index('"')]

print(deg)
print(minu)
print(sec)
```

```
37
46
26.2992
```

Loops and Conditionals

In [8]:

```python
""" Question
The Fizz Buzz challenge.

Write a program that prints the numbers from 1 to 100 and for multiples o
If it is divisible by both, print FizzBuzz.

So the output should be something like below
1, 2, Fizz, 4, Buzz, Fizz ... 13, 14, FizzBuzz, ...

Breaking down the problem further, we need to create for-loop with follow

If the number is a multiple of both 3 and 5 (i.e. 15), print FizzBuzz
If the number is multiple of 3, print Fizz
If the number is multiple of 5, print Buzz
Otherwise print the number
Hint: See the code cell below. Use the modulus operator % to check if a n

for x in range(1, 10):
    if x%2 == 0:
        print('{} is divisible by 2'.format(x))
    else:
        print('{} is not divisible by 2'.format(x))
"""

output = []

for num in range(1, 101):
    if num % 3 == 0 and num % 5 == 0:
        output.append("FizzBuzz")
    elif num % 3 == 0:
        output.append("Fizz")
    elif num % 5 == 0:
        output.append("Buzz")
    else:
        output.append(str(num))

print(", ".join(output))
```

```
1, 2, Fizz, 4, Buzz, Fizz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14, FizzBuzz, 1
6, 17, Fizz, 19, Buzz, Fizz, 22, 23, Fizz, Buzz, 26, Fizz, 28, 29, FizzBuz
z, 31, 32, Fizz, 34, Buzz, Fizz, 37, 38, Fizz, Buzz, 41, Fizz, 43, 44, Fiz
zBuzz, 46, 47, Fizz, 49, Buzz, Fizz, 52, 53, Fizz, Buzz, 56, Fizz, 58, 59,
FizzBuzz, 61, 62, Fizz, 64, Buzz, Fizz, 67, 68, Fizz, Buzz, 71, Fizz, 73,
74, FizzBuzz, 76, 77, Fizz, 79, Buzz, Fizz, 82, 83, Fizz, Buzz, 86, Fizz,
88, 89, FizzBuzz, 91, 92, Fizz, 94, Buzz, Fizz, 97, 98, Fizz, Buzz
```

Functions

In [9]:

```
""" Question
Given a coordinate string with value in degree, minutes and seconds, conv

def dms_to_decimal(degrees, minutes, seconds):
    if degrees < 0:
        result = degrees - minutes/60 - seconds/3600
    else:
        result = degrees + minutes/60 + seconds/3600
    return result

coordinate = '''37° 46' 26.2992"'''

# Add the code below to extract the parts from the coordinate string,
# call the function to convert to decimal degrees and print the result
# The expected answer is 37.773972
# Hint: Converting strings to numbers
# When you extract the parts from the coordinate string, they are strings
# You will need to use the built-in int() / float() functions to
# convert them to numbers
x = '25'
print(x, type(x))
y = int(x)
print(y, type(y))
"""

def dms_to_decimal(degrees, minutes, seconds):
    if degrees < 0:
        result = degrees - minutes/60 - seconds/3600
    else:
        result = degrees + minutes/60 + seconds/3600
    return result

coordinate = '37° 46\' 26.2992"'

deg = int(coordinate.split('°')[0])
minu = int(coordinate.split('°')[1].split("'")[0].strip())
sec = float(coordinate.split("'")[1].replace('"', '').strip())

decimal_degree = dms_to_decimal(deg, minu, sec)
print(decimal_degree)
```

```
37.773972
```